SheetVision: Optical Music Recognition for Sheet Music to MIDI Conversion

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***Abstract*—This paper presents SheetVision, a pipeline that converts printed sheet music images into playable MIDI files. The system combines classical computer vision with a convolutional neural network trained on synthetic data. The pipeline covers binarization, staff line detection and removal, symbol segmentation, CNN-based classification, pitch determination, and MIDI generation. The scope is limited to single-voice, treble-clef melodies in C major with clean printed notation. Results show that individual pipeline stages perform reliably within this scope, and the system produces correct MIDI output for simple melodies.**

***Index Terms*—optical music recognition, convolutional neural network, staff line detection, MIDI generation, synthetic dataset**

I. Introduction

Optical music recognition (OMR) is the process of extracting musical information from sheet music images, analogous to OCR for text. Unlike text, musical notation is two-dimensional: vertical position encodes pitch and horizontal position encodes time, making the problem inherently more complex.

SheetVision targets a well-defined subset of OMR: single-voice, treble-clef melodies with clean printed notation. The pipeline detects and removes staff lines, segments symbols via connected components, classifies them with a CNN trained on synthetic data, and maps position and label to pitch and duration for MIDI output. The system supports whole, half, quarter, and eighth notes along with common rest types. Beamed eighth notes and sixteenth notes are outside the current scope, as beaming introduces segmentation challenges that are left for future work.

II. Related Work

Calvo-Zaragoza et al. [1] survey the OMR field in *ACM Computing Surveys*, defining a common pipeline covering preprocessing, symbol detection, classification, and semantic reconstruction. Their taxonomy guided the overall structure of SheetVision.

Calvo-Zaragoza et al. [2] treat staff line removal as a supervised pixel classification problem, showing that learning-based approaches outperform traditional morphological methods. SheetVision uses a simpler projection-based approach, which is sufficient for clean printed scores.

Baro et al. [3] demonstrate CNN-based music symbol classification trained on synthetic data, achieving strong accuracy on isolated symbols. Their use of synthetic training data is similar to our approach of generating samples from music fonts rather than labeling real score images.

III. Methods

SheetVision processes a sheet music image through five sequential stages: preprocessing and staff detection, staff line removal, symbol segmentation, CNN classification, and pitch and MIDI generation.

A. Preprocessing and Staff Detection

The input image is converted to grayscale and binarized using Otsu's thresholding. Staff lines are detected via horizontal projection profiling: for each row, black pixels are counted, and rows exceeding 30% of the image width are marked as staff line rows. Consecutive marked rows are merged by taking their mean y-coordinate to produce single line positions.

Merged lines are grouped into staves of five by analyzing inter-line gaps. Lines with small, consistent gaps belong to the same staff; a gap exceeding twice the running average spacing starts a new group. Only groups of exactly five lines are kept as valid staves. For system grouping, the ratio of the largest to smallest gap between consecutive staves is computed. A ratio above 1.8 indicates a treble-bass pair; otherwise each staff is treated as an independent treble system.

B. Staff Line Removal

Staff lines are removed using conditional pixel deletion. For each black pixel on a detected staff line row, we check for black neighbors above and below, skipping over adjacent rows that are also staff line rows to handle multi-pixel thick lines. If no vertical neighbor exists outside the staff line region, the pixel belongs to the staff line and is set to white. If a neighbor is found, the pixel is part of a note stem or symbol and is preserved.

C. Symbol Segmentation

Connected components are extracted from the staff-removed image using OpenCV's `findContours`. Each blob is filtered through four criteria: size filtering removes noise blobs and

oversized regions; aspect ratio filtering removes bar lines; x-position filtering discards blobs in the leftmost 12% of the image, where clefs and time signatures appear; and y-position filtering removes blobs more than two staff spacings above or below the staff, eliminating chord labels and lyrics. Each surviving blob is assigned to the nearest staff by its y-center. Blobs assigned to bass clef staves are discarded. The output is a list of symbol dictionaries containing the cropped image, centroid, bounding box, staff index, and system index, sorted by system then x-position to reflect reading order.

D. CNN Classification

Each cropped symbol is classified by a CNN with a 64×64 grayscale input. The network consists of two double-convolution blocks (32 and 64 filters, 3×3 kernels) with batch normalization, ReLU activation, max pooling, and dropout. A fully connected classifier with a 256-node hidden layer produces a 7-class output covering whole notes, half notes, quarter notes, eighth notes, block rests, quarter rests, and eighth rests.

The network was trained on 896 synthetic images (128 per class) generated from Bravura and Leland music fonts with rotation, translation, and Gaussian noise augmentation. Training used an 80/20 train-validation split, Adam optimizer, cross-entropy loss, and 10 epochs.

E. Pitch Determination and MIDI Generation

Pitch is determined geometrically. For each classified note, the y-center is compared against the stored staff line coordinates for its assigned staff. A position index is computed as:

IV. Experiments and Results

A. Synthetic Dataset

We generated a balanced synthetic dataset of 896 images (128 per class) across seven symbol categories using two SMuFL-compatible music fonts: Bravura and Leland. Each symbol was rendered at four font sizes (330, 390, 450, 510) on a 600-pixel canvas to simulate varying print resolutions. Rendered symbols were cropped tightly, resized to 64×64 preserving aspect ratio, centered on a white canvas, and binarized. Light augmentations were applied including random rotation (±2 degrees), translation (±2 pixels), and Gaussian noise (standard deviation 1–4). A fixed random seed ensured reproducibility.

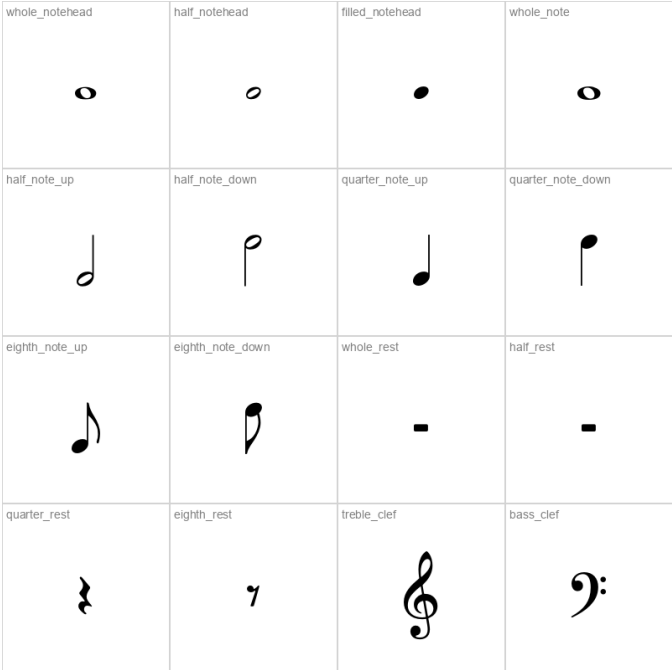

Fig. 1: Symbol preview rendered from the Bravura font.

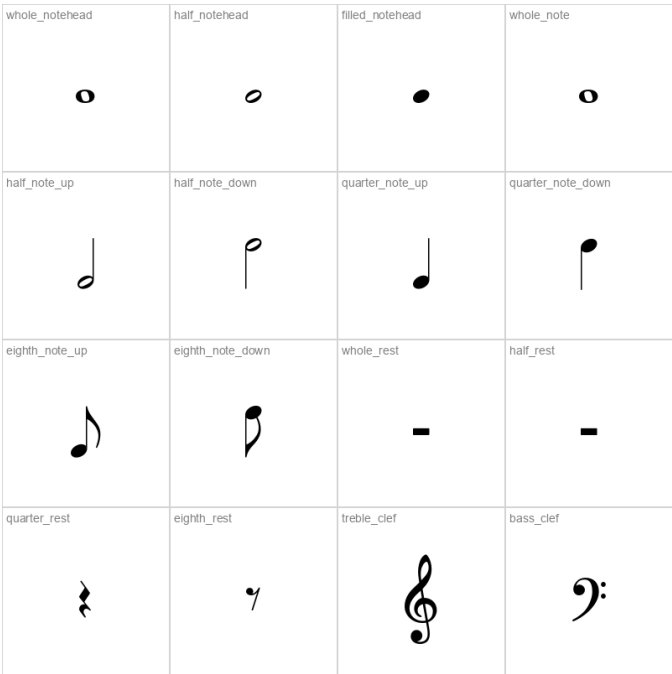

Fig. 2: Symbol preview rendered from the Leland font.

B. CNN Training

The CNN was trained on synthetic dataset using an 80/20 train/validation split, yielding 716 training samples and 180 validation samples. We used the Adam optimizer with a learning rate of 1×10^{-3} and a batch size of 32 over 10 epochs. Cross-entropy loss was used as the training objective. Input images were converted to grayscale, resized to 64×64, and normalized to $[-1, 1]$ prior to training. After 10 epochs, the model achieved a final training accuracy of **99.3%** and a validation accuracy of **100.0%**.

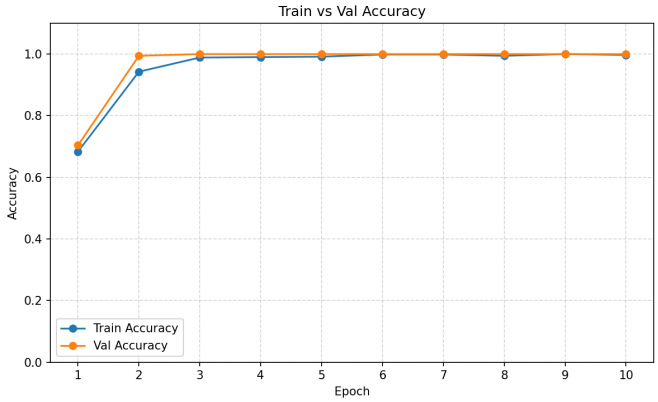

Fig. 3: Training and validation accuracy over 10 epochs.

C. Pipeline Evaluation

We evaluated the full pipeline on three treble-clef sheet music images: *Twinkle Twinkle Little Star*, *Jingle Bells*, and *Mary Had a Little Lamb*. Each image was processed end-to-end from raw input to MIDI output.

1) Staff Detection and Removal: Staff detection correctly identified all five-line staves across all test images, with consistent spacing values within each staff. Staff line removal preserved note stems and heads while cleanly removing horizontal lines.

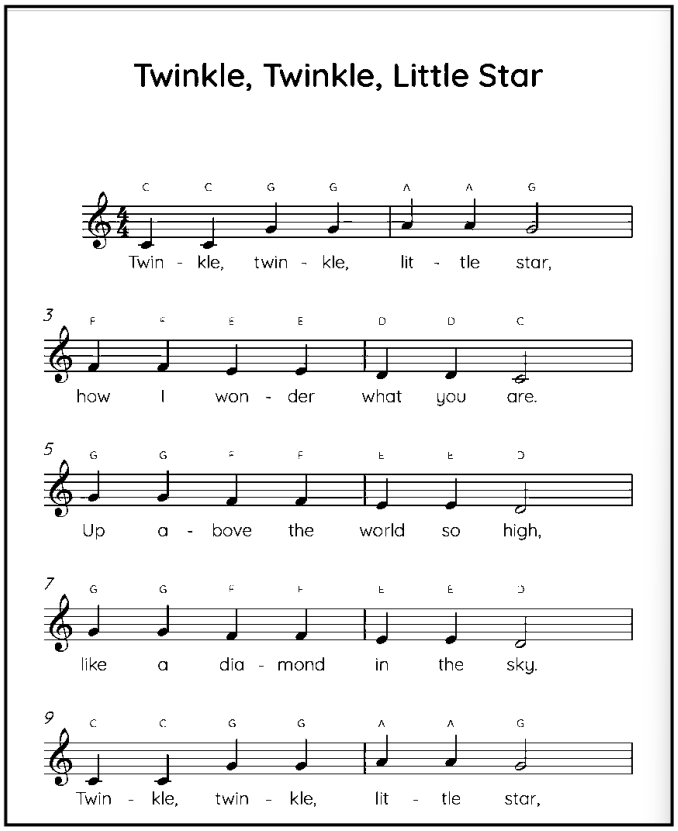

Fig. 4: Binarized input image.

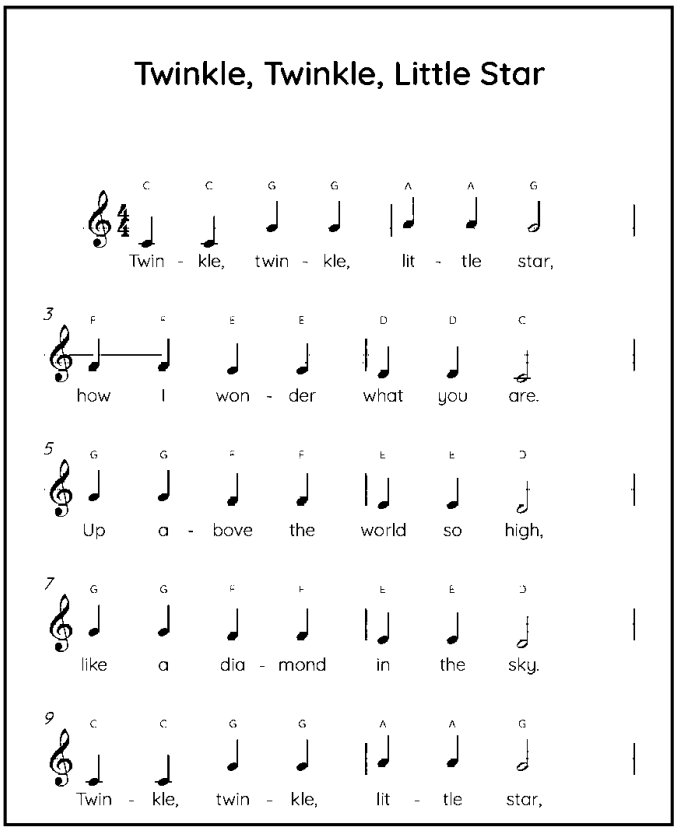

Fig. 5: Image after conditional staff line removal. Note stems and note heads are preserved while horizontal staff lines are removed.

2) Symbol Segmentation: Segmentation successfully isolated the majority of note symbols. The geometric filters effectively removed titles, lyrics, chord labels, bar lines, and clef symbols. A small number of false positives persisted, primarily clef fragments near the left edge and occasional staff line remnants.

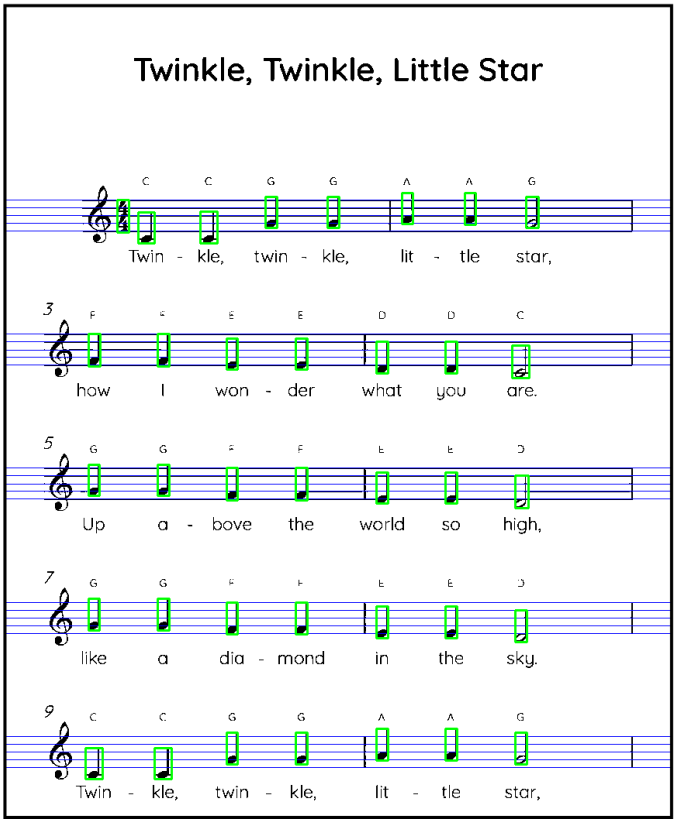

Fig. 6: Segmentation results with bounding boxes drawn around detected symbols. Staff line positions are shown in blue.

3) Classification and Pitch Accuracy: CNN classification achieved high confidence on correctly segmented symbols, with most notes classified above 99% confidence. Matching the inference preprocessing to the training data format (aspect-ratio-preserving resize, centering, and binarization) was critical for achieving this accuracy.

Pitch determination correctly identified the relative intervals between notes in all test pieces, producing recognizable melodies in the MIDI output. However, some notes were offset by one staff position due to the bounding box center not aligning precisely with the note head center, particularly for notes near the stem direction boundary.

4) MIDI Output: The generated MIDI files were playable and the melodies were identifiable across all three test pieces. Tempo was set to a default of 120 BPM. Duration classification was largely accurate, with quarter notes and half notes correctly distinguished in most cases.

V. Discussion and Summary

We presented SheetVision, an optical music recognition pipeline that converts printed sheet music into playable MIDI files by combining classical computer vision for preprocessing, staff detection, and segmentation with a CNN for symbol classification. The system performed well on simple treble-clef melodies, producing recognizable output for all three test pieces.

The main limitation is pitch accuracy. The bounding box center does not always align with the note head due to stem length and direction, causing occasional one-step pitch errors, particularly near the middle staff line where stem direction changes. A more targeted note head localization method, such as template matching or pixel density analysis within the bounding box, could address this. Segmentation also allowed occasional false positives through the geometric filters, which could be mitigated with CNN confidence thresholding.

The current scope handles single-voice treble-clef melodies with whole, half, and quarter notes. Natural extensions include beamed eighth note support (detecting note heads within wide connected components), bass clef processing (the pipeline already detects and labels bass staves), and accidental handling through key signature recognition. The modular pipeline design makes each of these extensions independently implementable without restructuring the existing system.

References

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[4] W3C Music Notation Community Group, "Standard Music Font Layout (SMuFL)," https://www.smufl.org/, 2023.

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